

Instagram User Analytics – SQL Project Report

Title Page

Project Title: Instagram User Analytics – SQL Fundamentals

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Tool Used: MySQL Workbench

1. Project Description

The goal of this project is to extract actionable insights from Instagram user data using SQL. As a data analyst, my task is to analyze how users interact with the Instagram platform and to answer specific business-related questions posed by the marketing and investor teams. These insights help guide strategic decisions around user engagement, campaign planning, and platform growth.

2. Approach

To tackle this project:

1. I loaded the provided Instagram database into MySQL Workbench.
 2. I wrote optimized SQL queries to analyze user behavior and engagement.
 3. I answered marketing and investor-specific questions based on the data.
 4. I compiled all queries, outputs, and insights into this report.
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3. Tech Stack Used

Tool	Purpose
MySQL Workbench	Writing and running SQL queries
SQL	Data extraction and analysis
MS Word / Google Docs	Report writing
Google Drive	File sharing

4. SQL Tasks & Queries

□ A) Marketing Analysis

1. Loyal User Reward – Oldest 5 Users

```
SELECT *  
FROM users  
ORDER BY created_at ASC  
LIMIT 5;
```

 **Insight:** These are the earliest adopters of the platform. They could be rewarded for loyalty.

2. Inactive Users – Never Posted a Photo

```
SELECT u.id, u.username  
FROM users u  
LEFT JOIN photos p ON u.id = p.user_id  
WHERE p.id IS NULL;
```

 **Insight:** These users could be targeted with email campaigns to encourage posting.

3. Contest Winner – Most Likes on a Single Photo

```
SELECT p.id AS photo_id, p.user_id, COUNT(l.id) AS total_likes  
FROM photos p  
JOIN likes l ON p.id = l.photo_id  
GROUP BY p.id  
ORDER BY total_likes DESC  
LIMIT 1;
```

 **Insight:** This user created the most engaging photo and is the winner.

4. Top 5 Hashtags

```
SELECT hashtag, COUNT(*) AS usage_count
FROM photo_hashtags
GROUP BY hashtag
ORDER BY usage_count DESC
LIMIT 5;
```

 **Insight:** These are the most popular hashtags to boost reach.

5. Best Day for Ad Campaign – Based on Registrations

```
SELECT
    DAYNAME(created_at) AS registration_day,
    COUNT(*) AS user_count
FROM users
GROUP BY registration_day
ORDER BY user_count DESC
LIMIT 1;
```

 **Insight:** Launch ads on the day with the highest registrations for better performance.

□ B) Investor Metrics

1. Average Posts Per User

```
SELECT
    COUNT(*) / (SELECT COUNT(*) FROM users) AS avg_posts_per_user
FROM photos;
```

 **Insight:** This metric shows how engaged the average user is.

2. Potential Bots – Liked Every Photo

```
SELECT l.user_id
FROM likes l
GROUP BY l.user_id
HAVING COUNT(*) = (SELECT COUNT(*) FROM photos);
```

 **Insight:** These users may be bots or automated accounts.

5. Insights

- A significant number of users never post; opportunity for marketing re-engagement.
 - A small subset of users drives most of the photo likes.
 - Certain hashtags dominate the platform—ideal for brand outreach.
 - User activity varies by day; the best day to launch campaigns can be optimized.
 - Potential bots can be flagged based on suspicious activity.
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6. Results & Learnings

- ✓ Completed full SQL-based data analysis project
 - ✓ Developed complex SQL queries to answer real-world business questions
 - ✓ Gained insights into Instagram user behavior
 - ✓ Identified growth opportunities for marketing and platform improvements
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